

ASHWINI AHER

• 716-563-9820 • aherashwini25@gmail.com • www.linkedin.com/in/ashwini-aher

EDUCATION

- **Master of Science, Electrical Engineering**, The State University of New York, USA GPA: 3.74 *Sept 2019*
- **Master of Engineering, Electronics and Telecommunication**, Fr. C. R. I. T., India GPA: 3.67 *Jan 2015*
- **Bachelor of Technology, Electronics and Telecommunication**, Dr. B. A. T. U., India GPA: 3.6 *June 2010*
 - **Relevant Courses:** MIMO Wireless Communications, Principles of Modern Digital Communication, Information Theory and Coding, Machine Learning, Deep Learning, Probability and Stochastic Process.

SKILLS

- **Programming:** Python 3, Julia
- **Tools:** QXDM, Wireshark, MATLAB, TensorFlow, Numpy, Matplotlib, Jupyter Notebooks
- **Concepts:** 5G architecture, NSA SA call flow, MIMO, OFDM, LTE, FDMA, TDMA, CDMA, WCDMA, OFDMA, SC-FDMA, Modulation Techniques, Source Coding, Channel Coding
- **Certifications:** Neural Networks and Deep Learning (Coursera)

RESEARCH EXPERIENCE

Multi-Kernel Recursive Least Squares in the Presence of Spares Outliers, with Dr. Slavakis, USA *Aug 2018-Aug 2019*

- Developed the “Outlier Robust Multi-Kernel Recursive Least Square (OR-MKRLS)” algorithm, using Julia.
- Concluded that the ‘OR-MKRLS with non-convex penalty’ outperforms the ‘OR-MKRLS with convex penalty’.

Hierarchical Clustering in Brain Network, with Dr. Slavakis, USA *Jan 2018-May 2018*

- Explored Hierarchical Clustering and its applications in Brain Networks.

Comparison of Vocal Tract Shape Estimation Based on Autocorrelation, Covariance, Lattice and Formant Frequencies. Dec 2014

- Utilized various Linear Predictive Coding (LPC) techniques to estimate Vocal Tract Shape using MATLAB.
- Published a research paper as a main author at ICNTE, India, (<http://ieeexplore.ieee.org/document/7029934/>).

WORK EXPERIENCE

Cellular System Test Engineer – Qualcomm *Nov 2022-Present*

- Continuing to utilize automation tools for modem software testing for cellular technologies such as 5G, LTE, WCDMA etc.
- Creating confluence page to make a note of high priority issues, using JIRA to track the issues.

Modem Integration and Test Engineer – Qualcomm *July 2022-Nov 2022*

- Tested numerous 5G, 4G features such as carrier aggregation, throughput for different band combinations, MO/MT voice calls, FDD/TDD scenarios, NR to NR Handover etc.

Modem Integration and Test Engineer – Novus Resources, Inc., Qualcomm *Nov 2020-July 2022*

- Worked with multi-discipline teams to integrate, test, verify and commercialize Qualcomm 5G technologies product.
- Tested modem software for various 5G NSA, 5G SA, 4G scenarios for worlds most advanced wireless modems.
- Performed log analysis using QXDM tool to debug the failures caught during testing.

Software Quality Analyst – BTI Solutions, Inc., Samsung Electronics America *July 2020-Nov 2020*

- Evaluated user interface software quality for SAMSUNG devices.
- Conducted different test cases related to 5G attach-detach, mobility, leg switch, EN-DC carrier aggregation, LTE only mode, throughput, VoLTE, different operators and their bands, different modes of tethering, MIMO, 64-QAM, 256-QAM, etc.
- Utilized diagnostic tools such as QXDM to collect logs, Umetrix Data for throughput and also collecting on-device logs.

Graduate Intern - FoodNerd, Buffalo, NY, USA *Feb 2019-May 2019*

- Created project proposal for Wearable Dietary Monetary Device project.

Executive Engineer - Steam Airetric Controls, India *Sept 2010-July 2012*

- Invented raw material management system for purchasing department and increased annual benefit almost by 99%.
- Led the production and purchasing departments and supervised effective working of them by scheduling weekly list of duties.
- Brought cost advantage equivalent to US\$10,000, by optimizing inhouse worker’s workload instead of giving it to contractors.

PROJECTS

Simulated MIMO system to find SER (MATLAB) *May 2018*

- Applied Jake’s fading simulator to generate Rayleigh fading channel for different number of channels. Furthermore, implemented MIMO systems based on Cyclic Space - Time Coding and Alamouti schemes.

Image recognition using deep Neural Networks (NN) (Python 3, Numpy) *Dec 2019*

- Trained Logistic regression, single layer NN and deep NN on ‘Cat vs non-Cat’ dataset with 12288 features. Deep neural network performance was best as compared to single layer neural network and logistic regression.